

USRP Studies

Week 1A: Overview of TCS and Research

8th Oct 2026

Overview

No crazy stuff; just an TCS and the practice of research in general. Following lecture 1 of the CS Theory Toolkit course @ CMU

1a. Scope of the Course

Note. For this course, TCS = Algorithms and Computational Complexity.

1b. Why Study TCS

Algorithms and complexity are present in every area of life, so you can study whatever you feel like studying (hyperbolically).

As a Researcher

- Regularly consume:
 - **Videos:** proceedings of recent FOCS/STOC conferences.
 - **Papers:** arXiv, ECCC. Check citing papers, since a given paper may not be the most recent result.
- Maintain a lifetime `.bib` file for all papers you ever reference.

1c. Tools for Research Problems

Based on the book *Street-Fighting Mathematics*, there are several approaches to doing research problems:

- **Inverse symbolic calculator:** finds expressions that evaluate to a given number (e.g. 3.14159265).
- **Mathematica/Maple:** plotting, testing numerical conjectures, etc. MATLAB for more numerical questions.